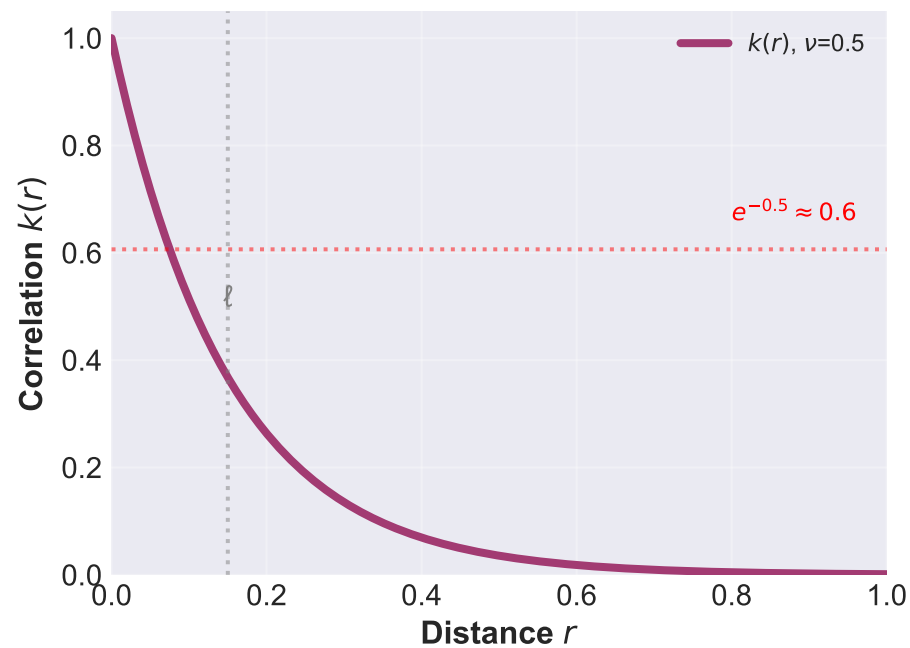
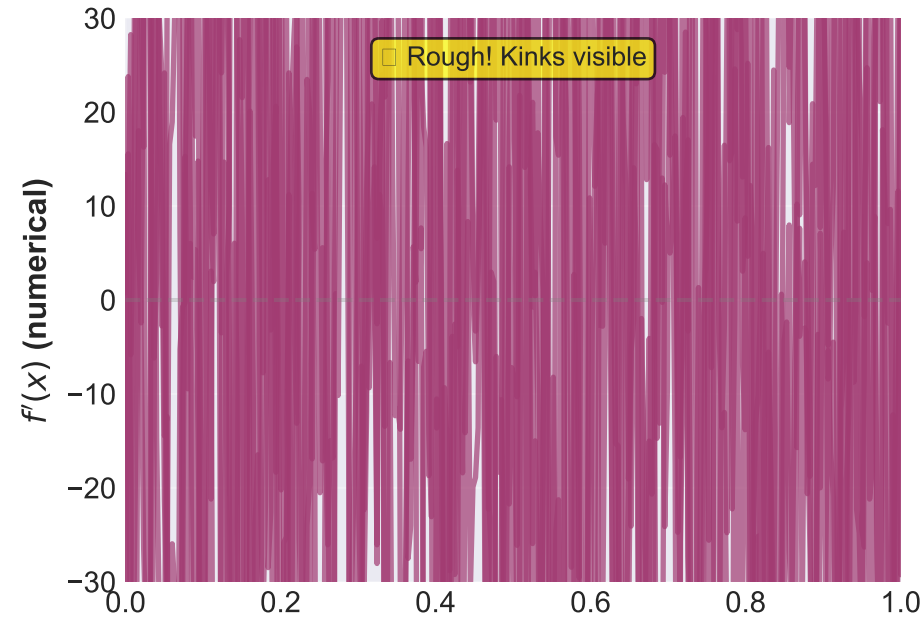
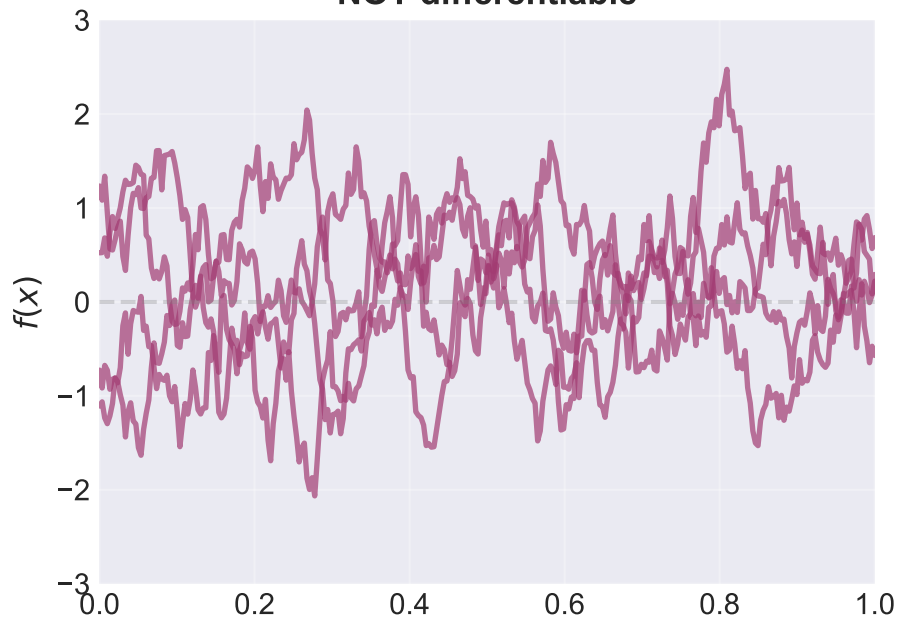
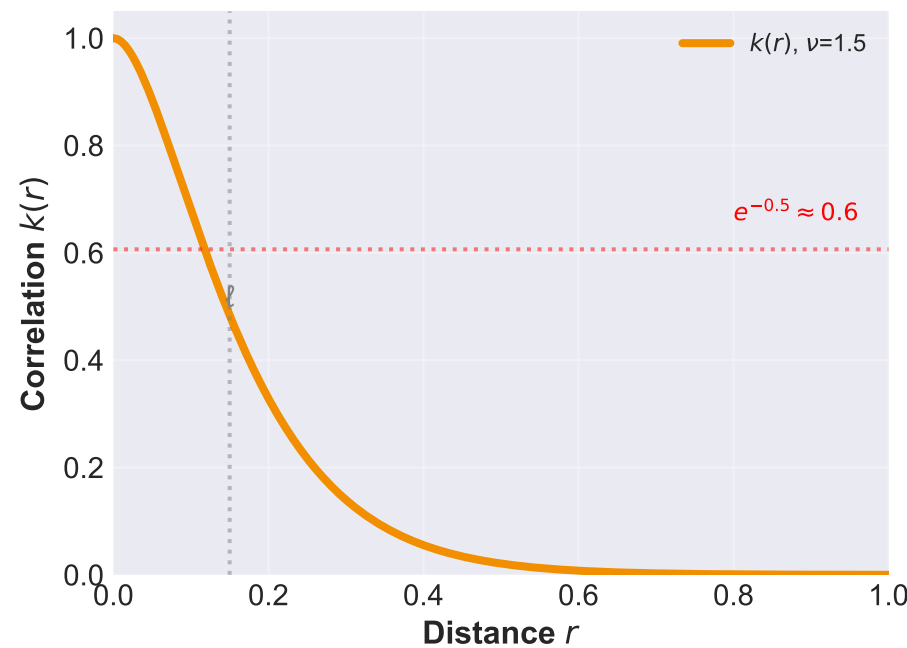
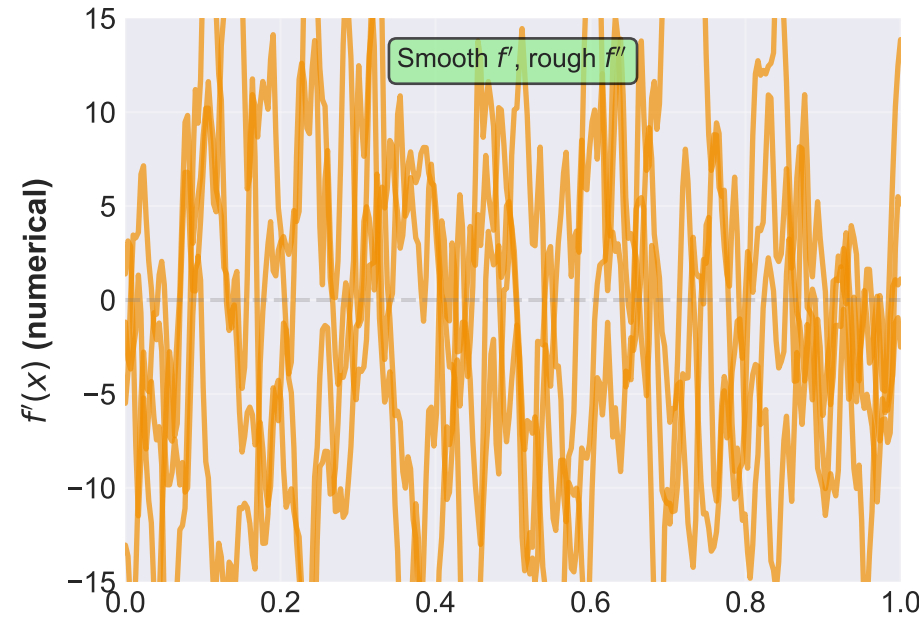
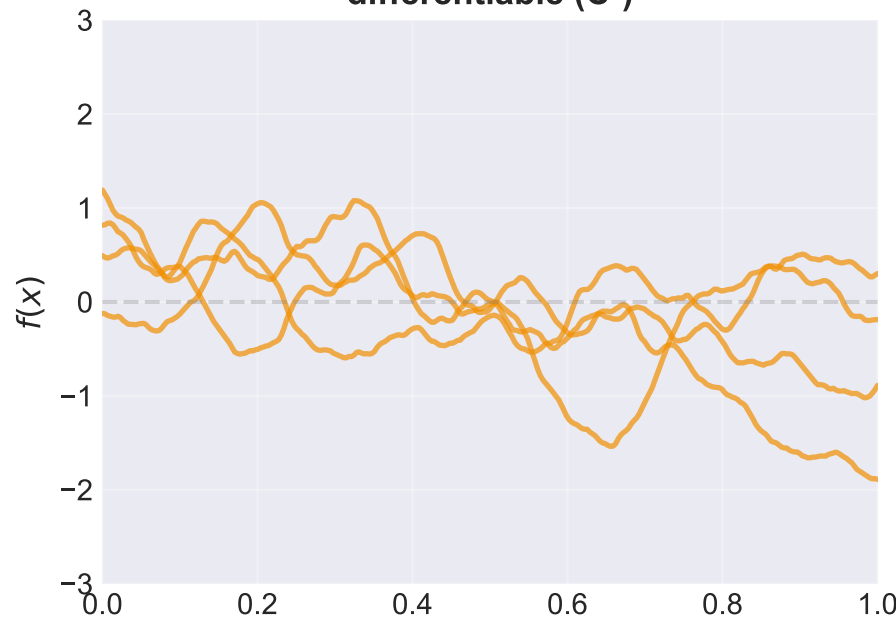


Matérn Smoothness Comparison: How ν Controls Differentiability

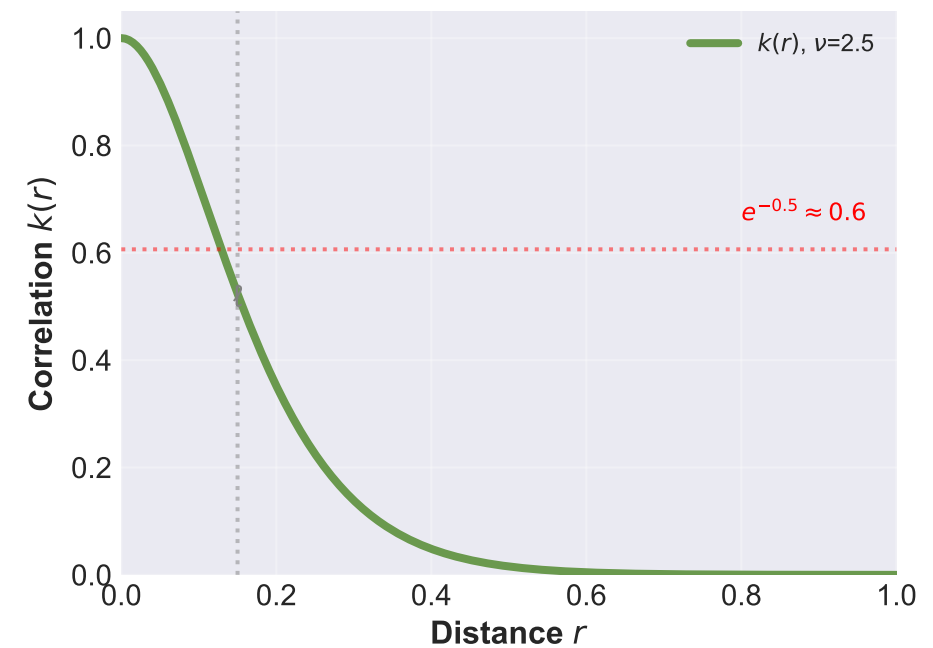
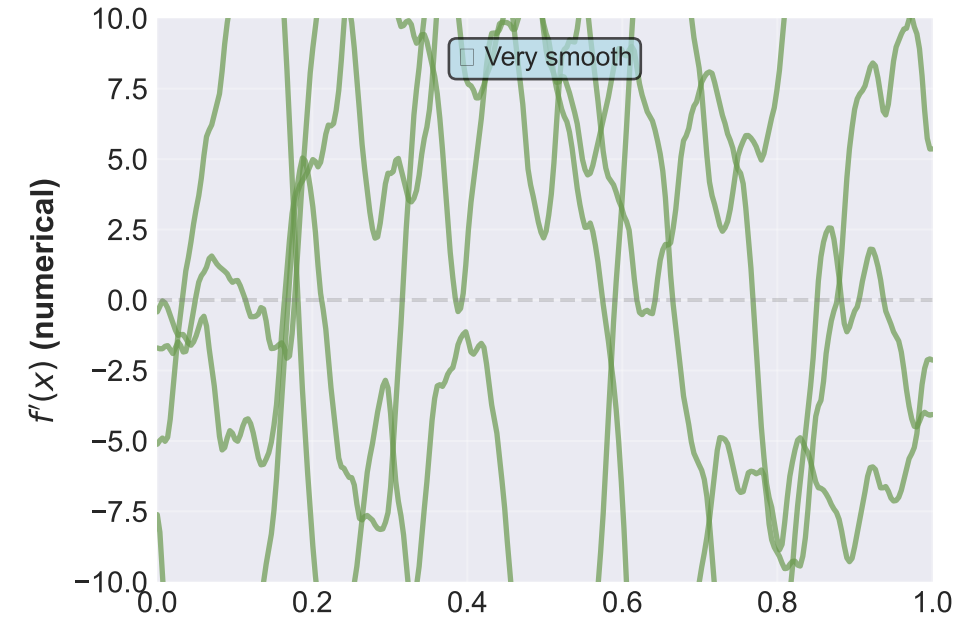
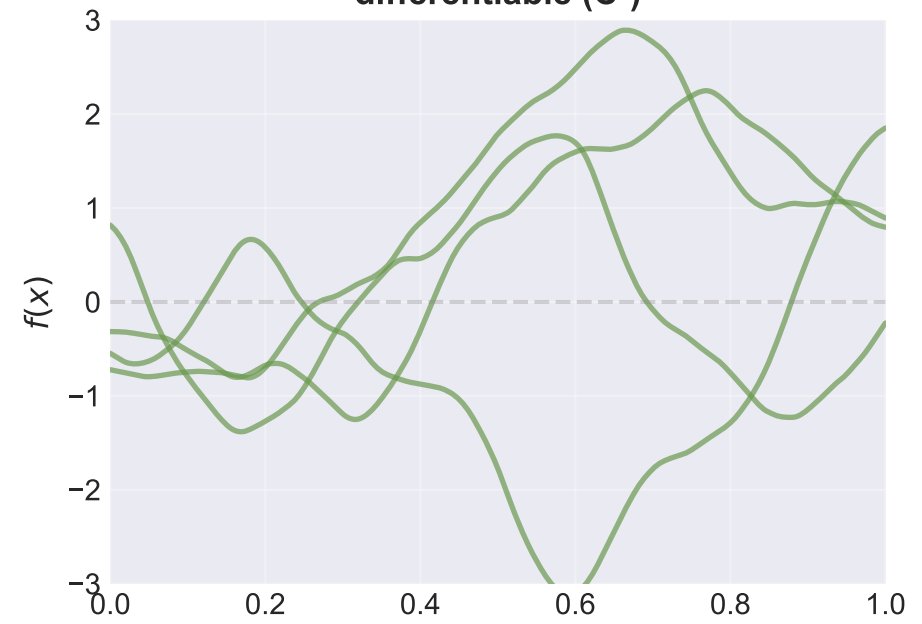
Matérn-1/2
 $\nu = 1/2$: Continuous
NOT differentiable



Matérn-3/2
 $\nu = 3/2$: Once
differentiable (C^1)



Matérn-5/2
 $\nu = 5/2$: Twice
differentiable (C^2)



Top row: Function samples $f(x)$. Middle row: Numerical derivatives $f'(x)$ reveal roughness—note how $\nu=1/2$ shows kinks (not differentiable), while $\nu=5/2$ is very smooth. Bottom row: Kernel correlation $k(r)$ shows how quickly correlations decay. Matérn-5/2 is recommended default: smooth enough for most physics, more realistic than infinitely-smooth SE.